

Scalability & Future Plan

Date	14 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The system currently provides crop recommendations based only on soil nutrients and environmental parameters.	Real-time weather conditions and market information are not considered.	High
2	The application does not maintain user accounts or prediction history.	Users cannot save or review previous recommendations.	Medium
3	The current version does not include fertilizer recommendations or disease detection.	Farmers receive only crop recommendations, limiting decision support.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Application is deployed on Vercel and accessible online.	Optimize the application and scale backend services to support a larger number of concurrent users.
2	Data Storage	Uses a CSV dataset and a Pickle (.pkl) model.	Integrate a cloud database to store user data and prediction history.
3	Performance	Predictions are generated using the trained machine learning model deployed with the application.	Optimize the model and improve response time for larger workloads.
4	Security	The application is publicly accessible without user authentication.	Add secure user login, role-based access, and HTTPS security enhancements.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Integrate real-time weather forecasting for more accurate crop recommendations.	Short Term	Improves prediction accuracy based on current weather conditions.
Phase 2	Add fertilizer recommendation based on soil nutrient levels.	Medium Term	Helps farmers choose suitable fertilizers and improve crop yield.
Phase 3	Implement disease detection using image processing and machine learning.	Medium Term	Enables early identification of crop diseases and reduces crop losses.
Phase 4	Add multilingual support and market price information for crops	Long Term	Improves accessibility for farmers and helps them make better farming and selling decisions.